



Game Design Document (GDD)

mdh

Mediadesign Hochschule

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1. Game Information

1.1. Game Concept

DouDash is a couch co-op 3D split-screen platformer set in an overgrown, Berlin-inspired city where nature has reclaimed its space. Players control a resourceful maintenance robot and a sentient plant with the power to manipulate greenery as they work together to solve intricate puzzles and overcome environmental obstacles. The game emphasizes teamwork, requiring players to combine their unique abilities - mechanical gadgets and control over plant life - to navigate the crumbling city and restore balance between nature and technology. Throughout the game, players follow the story of a robot determined to complete its task, with a curious plant that assists and accompanies it on this journey.

1.2. Experience Goals

The player experience goals of DouDash include delivering challenging and strategic gameplay that hinges on player cooperation. The game aims to provide exciting and satisfying jump-and-run mechanics, complemented by strong audiovisual appeal, ensuring a fun and engaging experience for all ages.

1.3. Features

- Couch Co-op Splitscreen
- Input Devices: Controller and/or keyboard & mouse
- Multiple Levels with progression system
- Parkour
- Teamwork required obstacles/puzzles
- Berlin inspired Landscape
- Semi realistic artstyle
- Immersive sound design
- Intuitive Inputs

2. Game References

The main Game Reference of DuoDash is 'Fireboy & Watergirl', which have to play as opposites (robot & plant) and therefore only able to interact with certain objects or touch certain surfaces. With the game being a Couch Co-op we decided to use Split Screen like seen in 'It takes two'. 'Kirby & the forgotten Land' as well as 'Overwatch' & 'Valorant' were used as Art inspiration for this game. 'Mario Odyssey' was used as a reference for the game's level and game design.

3. Gameplay

3.1. Game loops

The core game loop is that the players start a level, complete puzzles by platforming & working together and then finish the level by stepping on the goal platforms. When the players complete a level they unlock the next unless it's the last.

3.2. Objectives

In DouDash, the goal is for players to team up and solve puzzles together while overcoming obstacles to reach the end of each level. Success relies on cooperation, with both players combining their skills to tackle challenges in a visually immersive world. The game is designed to deliver challenging and strategic gameplay, with exciting and satisfying jump-and-run mechanics that keep the action engaging. With its strong audiovisual appeal, DouDash offers a fun and rewarding experience for players of all ages as they work side by side to complete each level.

3.3. Game Mechanics

- 3 different Versions of balancing platforms
- Moving Platforms
- Trampoline
- Vanishing & Killing Platforms
- Pulley
- Elevator
- Seesaw
- Pushing Players
- Buttons
- Switches
- Controllable Platforms

4. Characters

We have a robot and a plant as the playable characters. Both characters are the same size, can push, 'walk', run & jump, but they also have different abilities, like interacting with technology and hacking as a robot and interacting with nature and growing platforms as a plant.

5. Game World

5.1. Settings

General Setting is an overgrown & abandoned City with inspiration from the Capital of Germany, Berlin.

Information on gameelement and narrative are told via notes the players can collect.

5.2. Narrative

- Humans are still somewhere on earth, but abandoned the city for some reason
- Robot was left behind & activated by the humans to release signal that will destroy all plant life
- Plant could be a mutant/-ion (can't talk)
- Robot has a task, it's trying to fulfill (getting to Fernsehturm to release signal) and plant is following out of curiosity

6. Graphics

The game has midpoly models with a semi realistic art style, which is inspired by the looks of games like 'Overwatch' & 'Valorant'. The environment is filled with greenery to translate the look and feeling of an overgrown and abandoned city.

Each player sees the world through a third person's view.

The camera is within the parent_char blueprint. It is attached to a camera arm, which is following the player's skeletal mesh.

On jump the anchor of the camera on the character is detached on the z-axis.

The camera can be moved around the player in a sphere with the mouse or controller. The camera arm checks, if anything is between the camera and the player and if there is, moves closer to the character, until there is nothing in between.

7. Screen Display

DuoDash is a split screen present, since the game is a couch co-op.

Nothing is on the screen permanently present when playing the levels but we do have text that appears on the bottom of the whole screen.

Also when a player is in proximity with something to interact with a key prompt will appear on their side on the screen corresponding with what type of input device is connected with the character.

8. Multiplayer

The game is a 3D platform couch co-op and therefore a multiplayer only played on one device. We did not want to have a rigid camera the player cant influence. Also is that kind of system difficult to successfully implement without the player getting frustrated is a 3D world. With that in mind we decided to have splitscreen in our game like 'It takes two', so both players can control their field of view and level designers don't have to work with a new variable when building the map and jumps.

9. Development Softwares

a. Discord

Discord was used for the main communication outside from seeing the people of the team in person.

Meetings were held through discord as well, like the art and Sunday meetings. It was also used to report bugs, extra version control, announcements to meetings or other important events but it also had channels which were used to have fun (for example our memes and tea channels).

b. Miro

Miro had in our case 3 boards: Art & level/gamedesign, general and idea dump.

General was used to keep track of tasks each person has to do that week and was 90% maintained by each person themselves. If tasks were missing or if it was evident that it was not maintained the producers gave strict reminders what each person's tasks are this week. Each week the color of the tasks changes, which makes it easier to see if each team member updated their tasks on miro.

Art & level/Game Design was mainly used to show the process of the artist and to have a place to show their work and easily discuss it.

Idea dump is a board used to gather ideas and was used a lot in the beginning of the project. Because discussing ideas with everybody will take up a whole day, the ideas of that board were collected, organized, sorted (useful & not doable, ...) and then those selected were discussed with everybody or a selected group of people.

c. Perforce

Perforce was used instead of a tool like github, due to its feature to lock files you are working on to minimize the problem of version control and errors in code.

We did not only use perforce to contain the Unreal project but als to share documents or assets that still need to be worked on by another person before implementing it into the engine.

d. Codecks

Codecks was used to keep track of all tasks similar to Jira. Each Lead or a specific person was responsible for keeping the decks up to date and organized.

e. Google Docs

Google Docs were used to do all the documentation for the team, due to it being easy to share and edit or see the edits each person did.

Those documents were also used to keep all people on the team up to date, when they're not able to participate in meetings or the general stand of the project.

f. Unreal Engine

Unreal Engine 5 is the Engine used to create the Game in with the Version 5.4. We changed versions in the beginning frequently due to earlier versions having problems with the split screen. Even though Unreal Engine 5 can be used with C++ to create the project, we decided on doing it solely with blueprints, due to all programmers having no experience working in a game-engine and the second semesters feeling more comfortable with blueprints.

g. Blender

Blender was the Software used to create all 3D assets of the game.

h. Adobe Photoshop/Illustrator

Both Adobe Photoshop as well as Illustrator are used to create the UI assets as well as Decals, such as cracks or graffiti.

i. Ocenaudio

Ocenaudio is used to edit and adjust the sound files, which are then later added to the engine.